IIMT3642: Managing and Mining Big Data - Classification (Decision Tree III)

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Lecture Outline

- Recap
- A simple measure for evaluating classification performance
- Overfitting and Generalization
- Decision Making with Decision Tree Model

Reading for today's lecture

Data Science for Business, Foster Provost and Tom Fawcett, O'Reilly, 2013.

pp. 111 – 118, 126-130, 133-136, 71-79

Recap

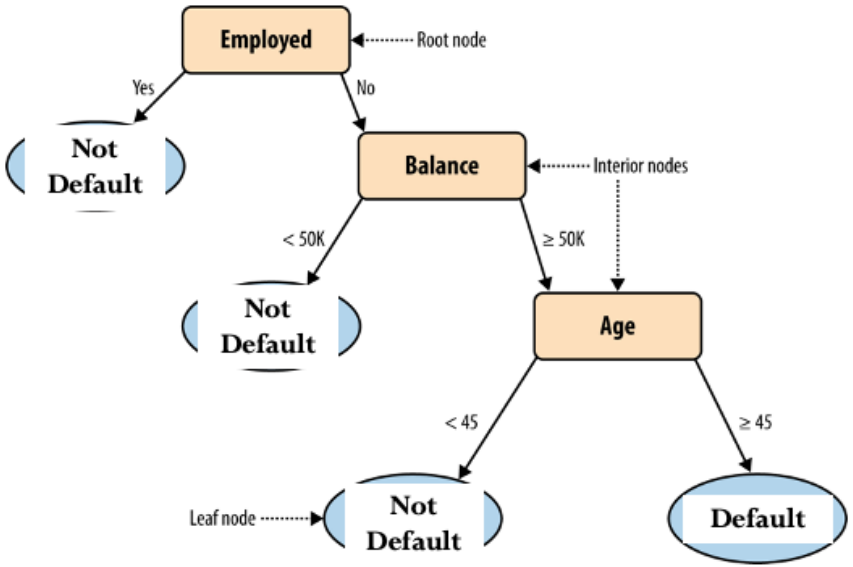
- Classification is the process of learning a model (from training data) that can assign an instance of interest (whose class is unknown) to one of the predefined classes.
  - A classification model is a predictive model because the class of an instance to be classified is unknown.
  - For example, we want to predict whether a customer belongs to the “Churn” class or the “Not Churn” class.

- Training data consist of instances whose class is known. Each instance is described with descriptive attributes and a class label attribute.

Name	Balance	Age	Employed	Default
Mike	$200,000	42	No	Yes
Mary	$350,00	33	Yes	No
Claudio	$115,000	40	No	No
Robert	$29,000	23	Yes	Yes
Dora	$72,000	31	No	No

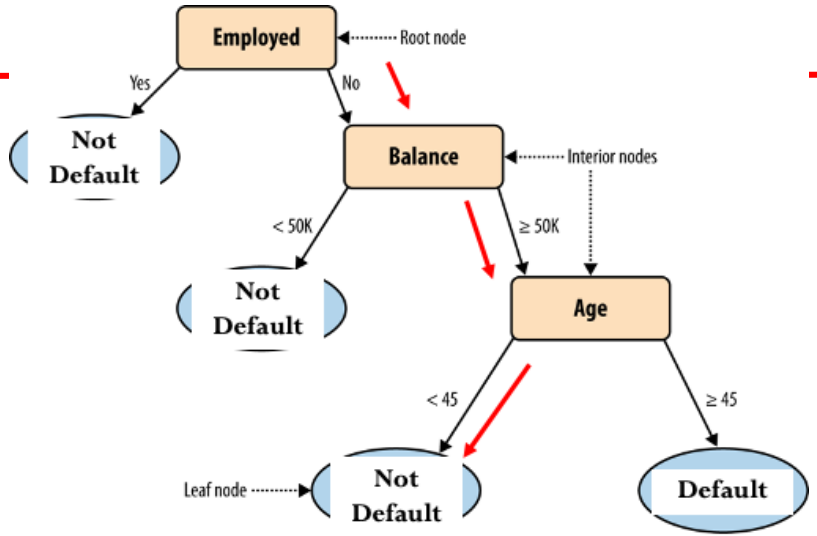
The classification process consists of two steps:

- Step 1: Learn a classification model from training data;
- Step 2: Apply the model to classify an instance with unknown class.



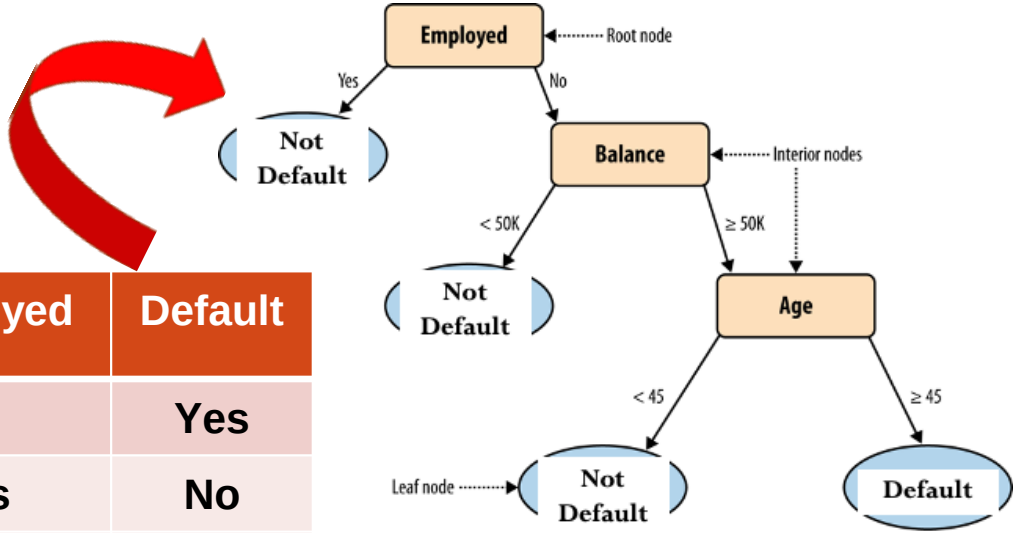
- Each internal node or the root node represents a test of a descriptive attribute and the test result determines which branch to go.

- Each leaf node represents a class.



Name	Balance	Age	Employed	Default
Jay	$120,000	42	No	?

Name	Balance	Age	Employed	Default
Mike	$200,000	42	No	Yes
Mary	$350,00	33	Yes	No
Claudio	$115,000	40	No	No
Robert	$29,000	23	Yes	Yes
Dora	$72,000	31	No	No



Recap: impurity measure

Entropy of a data set:

$$Entropy = -p_1 \times \log_2 p_1 - p_2 \times \log_2 p_2 - \cdots - p_k \times \log_2 p_k$$

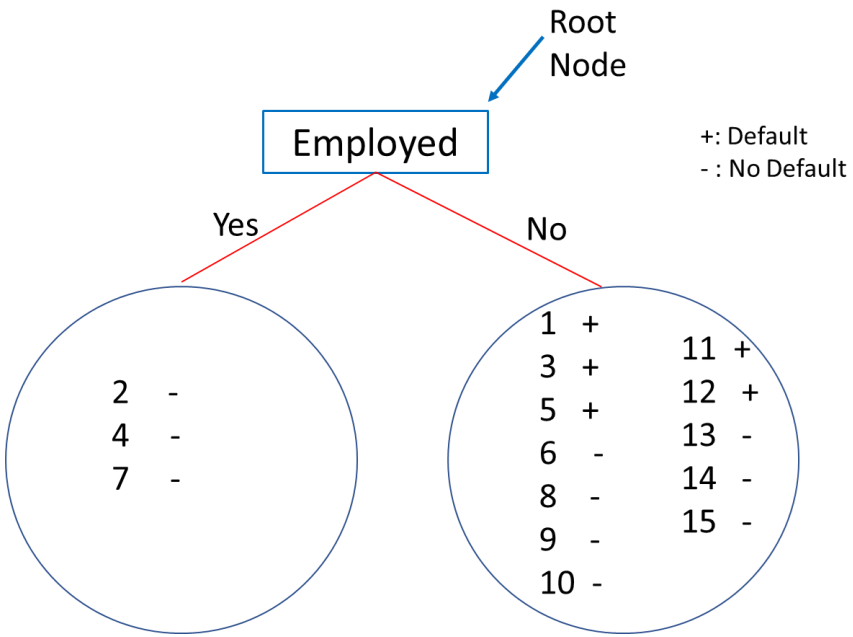
k: number of predefined classes

p_i : fraction of instances belonging to the i-th class

Note: $0 \times \log_2 0 = 0$

- Pick the attribute that gives the largest entropy reduction:

Attribute	Entropy Reduction
Employed	0.134
Age	0.026
Balance	0.03


Root
Node
Employed
+: Default
- : No Default
Yes
No
2 -
4 -
7 -
1 +
3 +
5 +
6 -
8 -
9 -
10 -
11 +
12 +
13 -
14 -
15 -

Recap: Decision Tree Learning

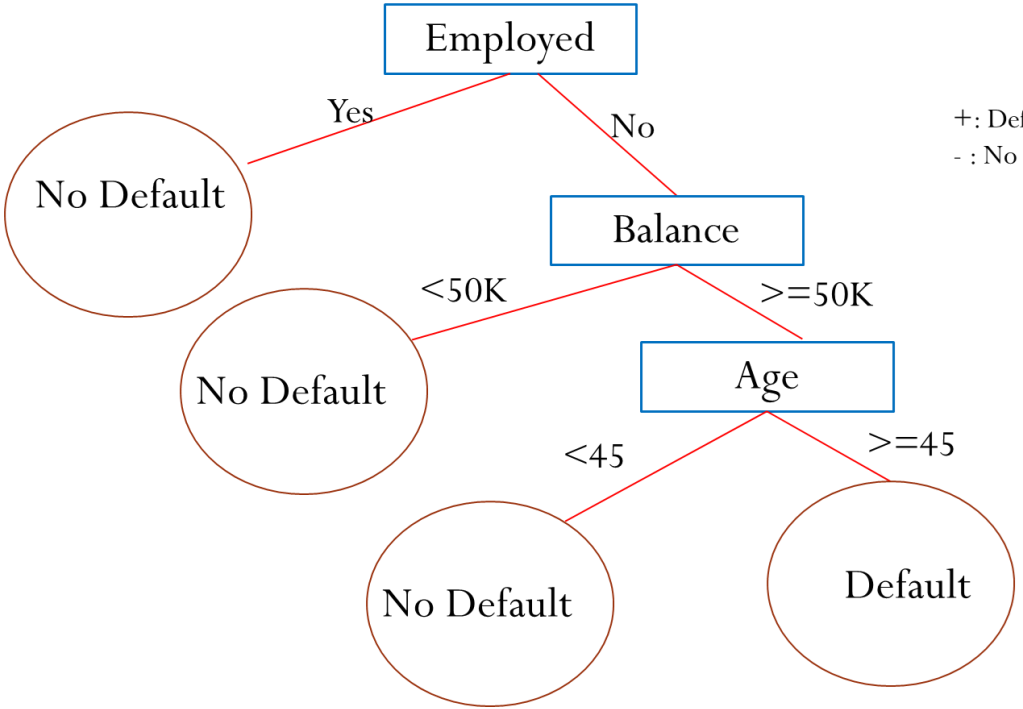
- Continue picking a descriptive attribute to divide each of the resulting data set unless
  - All records in a resulting data set are of the same class (No need to divide) OR
  - All records in a resulting data set have the same descriptive attribute values. (Cannot divide)
- If a resulting data set do not need to be further divided (or cannot be further divided), label it using the majority class among the records of the data set.

Measure Classification Performance

- An instance is associated with
  - actual class label
  - predicted class label: one predicted by a classification model (e.g., decision tree)
- An instance is correctly classified if
  - actual class label = predicted class label

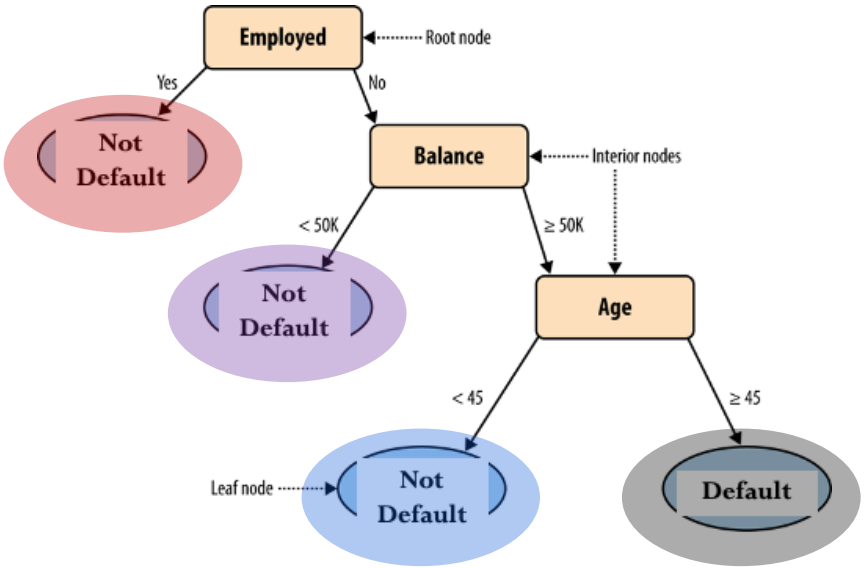
A Simple Measure

$$Accuracy = \frac{\text{number of correctly classified instances}}{\text{number of instances classified}}$$



+: Default
- : No Default

CID	Balance	Age	Employed	Default
1	<50K	>=45	No	Yes
2	<50K	<45	Yes	No
3	>50K	>=45	No	Yes
4	>=50K	>=45	Yes	No
5	>50K	>=45	No	Yes
6	<50K	>=45	No	No
7	>=50K	<45	Yes	No
8	<50K	>=45	No	No
9	<50K	>=45	No	No
10	>=50K	<45	No	No
11	>=50K	<45	No	Yes
12	>=50K	<45	No	Yes
13	>=50K	<45	No	No
14	>=50K	<45	No	No
15	>=50K	<45	No	No



CID	Balance	Age	Employed	Default	Predicted Default
1	<50K	>=45	No	Yes	No
2	<50K	<45	Yes	No	No
3	>50K	>=45	No	Yes	Yes
4	>=50K	>=45	Yes	No	No
5	>50K	>=45	No	Yes	Yes
6	<50K	>=45	No	No	No
7	>=50K	<45	Yes	No	No
8	<50K	>=45	No	No	No
9	<50K	>=45	No	No	No
10	>=50K	<45	No	No	No
11	>=50K	<45	No	Yes	No
12	>=50K	<45	No	Yes	No
13	>=50K	<45	No	No	No
14	>=50K	<45	No	No	No
15	>=50K	<45	No	No	No

$$Accuracy = \frac{12}{15} = 0.8$$

- A more advanced measure: Confusion Matrix
  - true class label: p (positive) n (negative)
  - predicted class label: Y (predicted positive) N (predicted negative)

True Class

Predicted Class

	p	n
Y	True Positive	False Positive
N	False Negative	True Negative

- True positive (TP): number of instances predicted as positive and actually positive.
- False positive (FP): number of instances predicted as positive but actually negative.
- True Negative (TN): number of instances predicted as negative and actually negative.
- False Negative (FN): number of instances predicted as negative but actually positive.

True Class

Predicted Class

	p	n
Y	True Positive	False Positive
N	False Negative	True Negative

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} = \frac{10+2}{2+10+0+3} = \frac{12}{15}$$

$$Precision(Yes) = \frac{TP}{TP+FP} = \frac{2}{2+0} = 1$$

$$Recall(Yes) = \frac{TP}{TP+FN} = \frac{2}{2+3} = \frac{2}{5}$$

$$Precision(No) = \frac{TN}{TN+FN} = \frac{10}{10+3} = \frac{10}{13}$$

$$Recall(No) = \frac{TN}{FP+TN} = \frac{10}{10+0} = 1$$

	p	n
Y	2	0
N	3	10

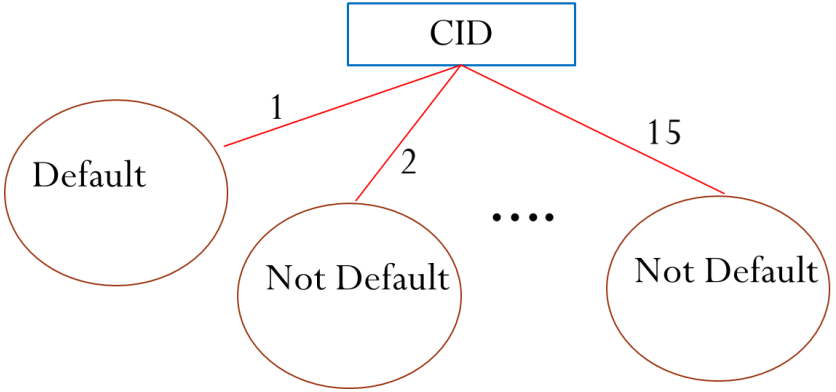
$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Are there any problem of using training data to evaluate the performance of a classification model learned from the same training data?

Overfitting and Generalization

- The objective of classification is to assign an instance of interest (whose class is unknown) to one of the predefined classes, NOT to predict class labels for instances in training data.
- A classification model with good (even perfect) accuracy for training instances may NOT predict well for instances whose classes are unknown.
- In other words, training accuracy is not a good indicator of a classification model's accuracy in predicting instances with unknown classes.

Given a hypothetical model:



- Training accuracy = 100% (perfect on the training data!)

How do you think this model will perform on an instance with unknown class label?

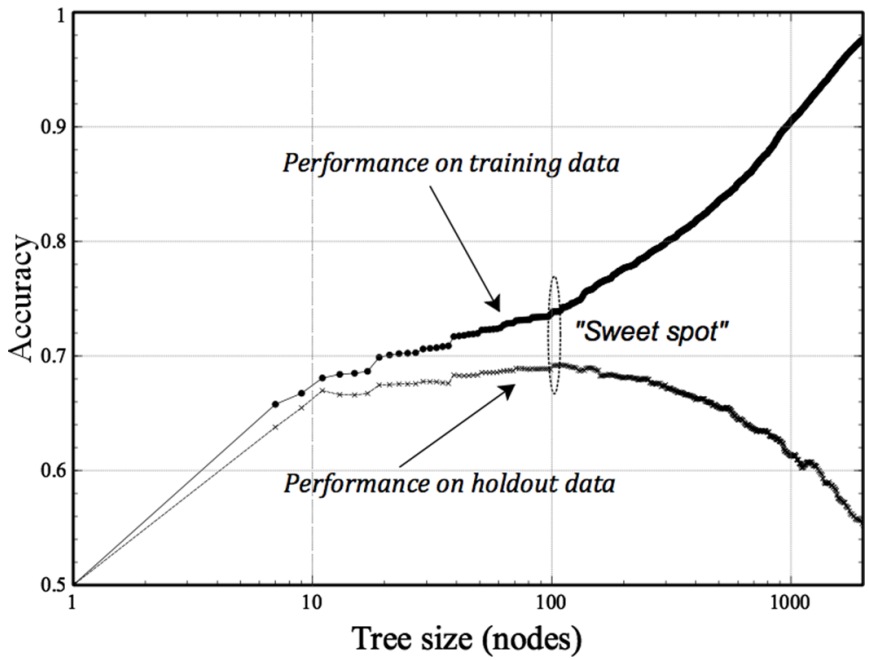
- Generalization is the property of a model, whereby the model applies to data that were not used to build the model.
- Overfitting is the tendency of tailoring a model to the training data, at the expense of generalization.
- As a model becomes more complex (e.g., more nodes in a decision tree), it becomes more likely to overfit.

Evaluation Data Split

- Training data: used to build the model; class labels of its instances are known to the learning algorithm.
- Holdout data (Test data): Not used to build the model; so that class labels of these instances are Unknow to the learning algorithm.

Commonly observed pattern during the model learning

- Performance on hold data diverges from that of training data, when the model becomes more complex
- There exists a sweet spot, when the model trained performs best on the holdout data



- Why overfitting: a model is so tailored that it begins to fit outliers and noises in training data, thus generalizes poorly.

- How to avoid overfitting— a perspective of using validation during the learning process:
  - Divide the known data set at hand into two parts: training and validation
  - Learn a model from training data
  - Evaluate the model using validation data
  - Keep track of their respective performances
  - Stop increasing model complexity when the gap between training and testing performances begins to widen

CID	Balance	Age	Employed	Default
1	<50K	>=45	No	**Yes**
2	<50K	<45	Yes	No
....	...	...	...	...
14	>=50K	<45	No	No
15	>=50K	<45	No	No

Training

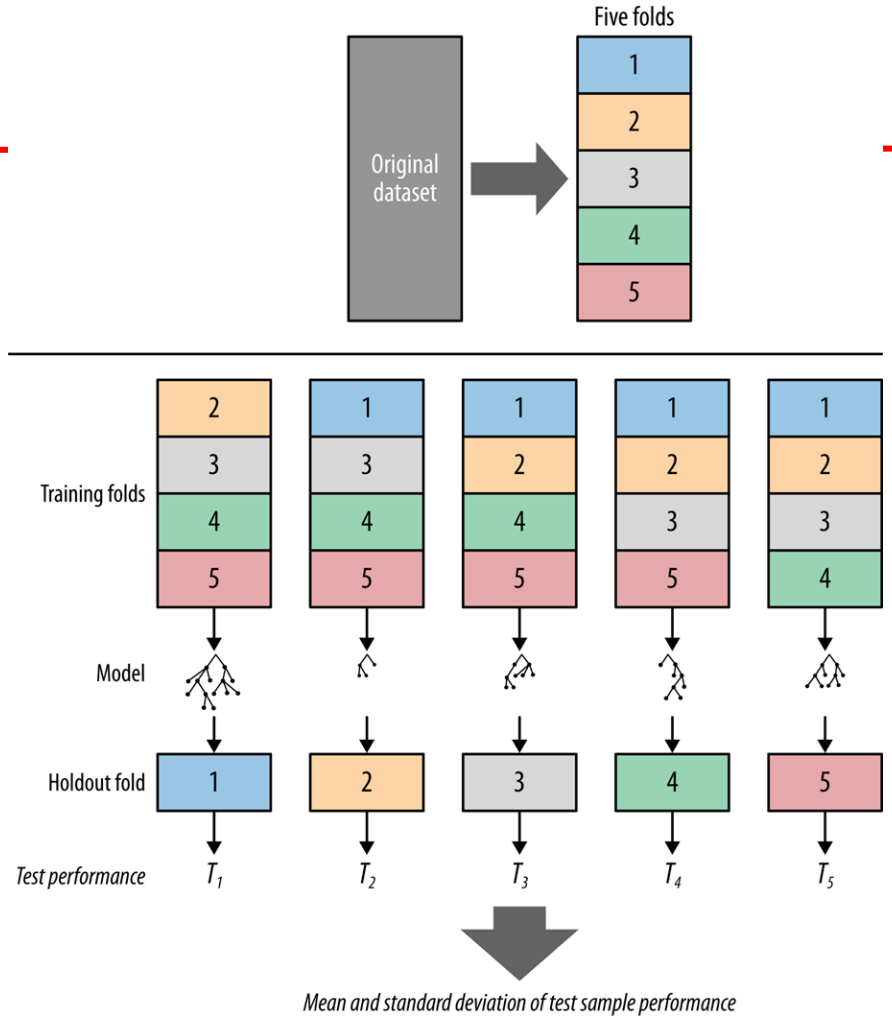
CID	Balance	Age	Employed	Default
1	<50K	>=45	No	**Yes**
2	<50K	<45	Yes	No
....	...	...	...	...
7	>=50K	<45	Yes	No

Validation

CID	Balance	Age	Employed	Default
8	<50K	>=45	No	**No**
....	...	...	...	...
14	>=50K	<45	No	No
15	>=50K	<45	No	No

- *K*-fold cross-validation

  1. Divide data into *K* equal-sized partitions
  2. Use each partition as test data and the rest partitions as training data
  3. Output *K* accuracies
  4. Typical $K = 5, 10$


Five folds
Original dataset
1
2
3
4
5
Training folds
Model
Holdout fold
Test performance
T_1
T_2
T_3
T_4
T_5
Mean and standard deviation of test sample performance

How to avoid overfitting — a methodological perspective

- pre-pruning: stop growing a decision tree before it gets too complex by adding a stop condition – minimum number of instances in a leaf node.
- post-pruning: grow a full-sized tree and then prune it (i.e., replacing branches with leafs)

Making Decisions with Decision Tree Model

- Decision rules
  - Decision rules can be generated by tracing down a path from the root node to a leaf
    - IF Employed THEN No Default
    - IF Not Employed AND Balance < 50K THEN Not Default
    - IF Not Employed AND Balance >= 50K AND Age <45 THEN Not Default
    - IF Not Employed AND Balance >= 50K AND Age >=45 THEN Default

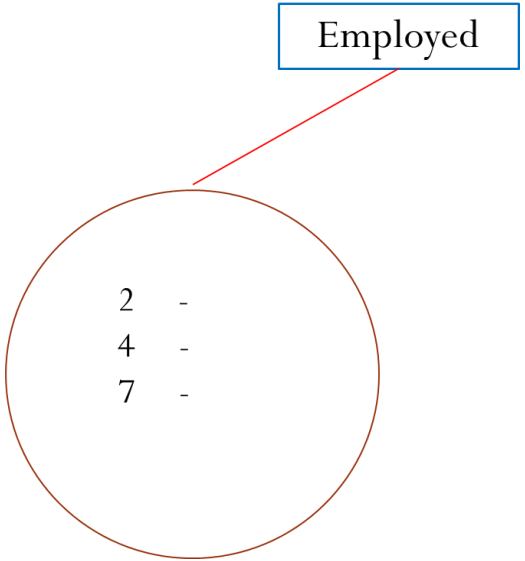
- Decision rules
  - Adding probabilities to decision rules

$$p(c) = \frac{\#\ of\ instances\ belonging\ to\ class\ c}{\#\ of\ instances\ in\ the\ leaf\ node\ corresponding\ to\ a\ rule}$$

- Decision rules
  - Adding probabilities to decision rules
  - Example #1: the branch IF Employed

If Employed

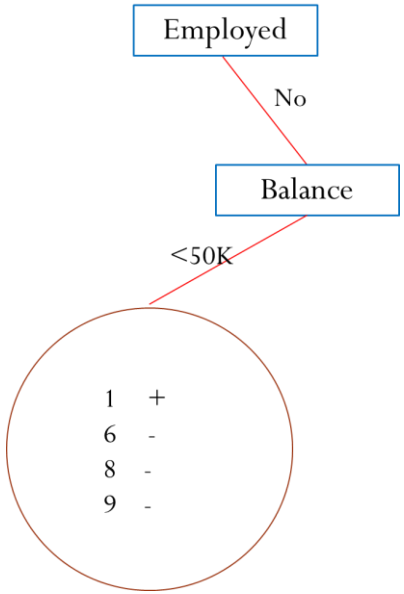
then p(No Default) = 3/3=1 and p(Default) = 0/3 = 0



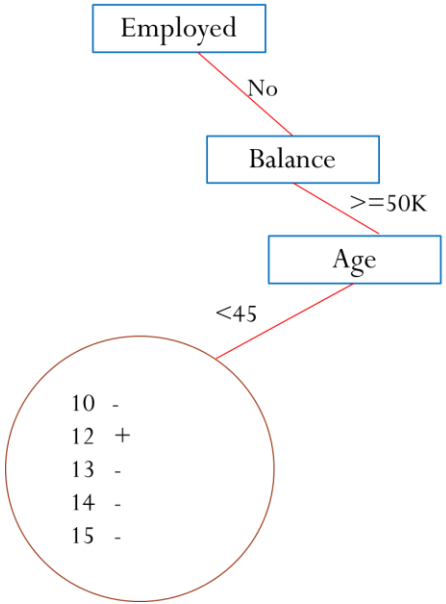
- Decision rules
  - Adding probabilities to decision rules
  - Example #2: the branch IF Not Employed and Balance < 50k

If Not Employed and Balance < 50k

then p(No Default) = 3/4 =0.75 and p(Default) = 1/4 = 0.25



- Decision rules
  - Adding probabilities to decision rules
  - Example #3: the branch IF Not Employed and Balance >= 50k and Age < 45

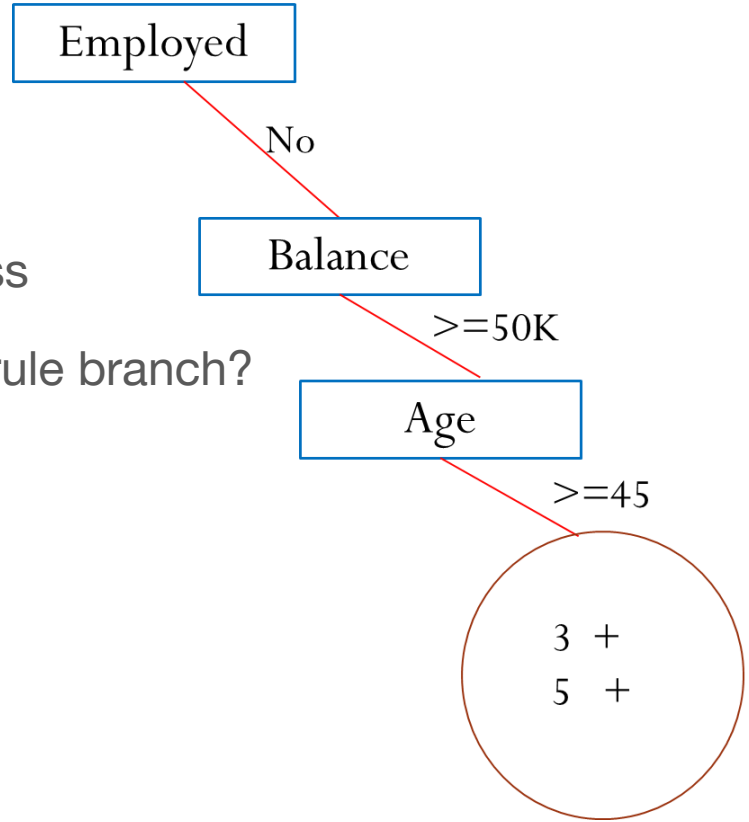


If Not Employed and Balance < 50k and Age < 45

then p(No Default) = 4/5 =0.8 and p(Default) = 1/5 = 0.2

What is the expected probability of each class

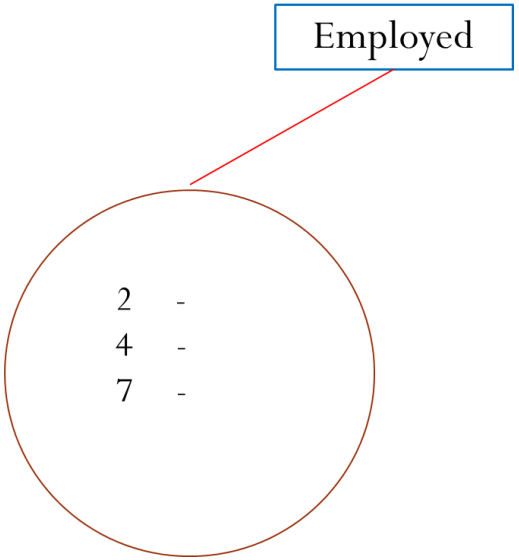
for an instance that falls under this decision rule branch?



- Laplace correction

$$p(c) = \frac{\#\,\text{of instances belonging to class C} + 1}{\#\,\text{of instances in the leaf node corresponding to a rule} + \text{K}}$$

K: number of predefined classes



If Employed then $p(No\ Default) = \frac{3+1}{3+2} = 0.8$ and $p(Default) = \frac{0+1}{3+2} = 0.2$